MILITARY SPECIFICATION

PROJECTILE, 165MM, TP
M624, METAL PARTS

This specification is approved for use by the US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers Projectile, 165MM, TP, M624 Metal Parts.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-F-320	Fiberboard, Sheet Stock and Cut Shapes
PPP-B-636	Box, Fiberboard

MILITARY

MIL-I-6868	Inspection Process, Magnetic Particle
MIL-A-48078	Ammunition, Standard Quality Assurance Provisions, General Specification For
MIL-C-50413	Cartridge, 165MM, TP, M623 Loading, Assembly and Packing

FSC: 1320

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-410	Nondestructive Testing Personnel Qualification and Certification
MIL-STD-1169	Packaging, Packing and Marking of Inert Ammunition Components

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

10543024	Projectile, 165MM, TP, M624, Metal Parts
8837835	Pallet for Fixed and Semifixed Shells
9219045	Cartridge, 165MM, TP, M623, LAP

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwis indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ANSI - B46.1 Surface Texture

(Application for copies should be addressed to the American Nationa Standard Institute (ANSI), 1430 Broadway, New York, N.Y. 10018).

ASTM E8 Tension Testing of Metallic Materials
ASTM B117 Salt Spray (Fog) Testing

(Application for copies should be addressed to the American Society for Testing and Materials (ASTM), 1916 Race Street, Philadelphia, Pa. 19103).

3. Requirements

3.1 General.-Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings and reference specifications.

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3.2 Stress relief.-Projectile bodies shall be stress-relieved at a minimum temperature of 750 degrees Fahrenheit, for a period of one hour, minimum, following the completion of all cold-work operations. Stress-relief shall be uniform throughout and shall be so conducted as to produce the mechanical properties specified on the applicable drawing.

3.3 Mechanical properties.-The mechanical properties specified on the applicable drawings shall apply for the projectile body and the base plug.

3.4 Surface preparation and application of protective finish.-Where cleaning and surface pretreatment are required by component details, the process shall apply to all indicated surfaces regardless of contour configuration. The protective finish shall be as specified on the applicable drawings and shall be uniformly applied to the surfaces designated by the applicable drawings. Salt spray and other required testing shall be in accordance with the applicable drawings and specifications.

3.5 Defective raw material and metal parts components

3.5.1 Billet.-The billet shall be free from cracks, splits, cold shuts, inclusions, pipe, porosity, folds, bursts, seams and other metal defects.

3.5.2 Projectile body.-The projectile body shall be free from seams, cracks, splits, stingers, laps, folds, slugs, porosity, dents, scale, fins, draw marks, laminations, pits, sharp edges, and imbedded foreign matter.

3.5.3 Base plug.-The base plug shall be free from cracks, folds or other metal defects.

3.5.4 Visual standards.-Visual standards depicting maximum acceptable degrees of conditions described in 3.5.1, 3.5.2 and 3.5.3 will be established for each contract after the first month's or lot's production, whichever quantity is smaller. In the absence of a visual standard, the requirements of 3.5.1, 3.5.2 and 3.5.3 shall apply (see 6.5).

3.6 Proving ground tests.-The assembled projectile metal parts shall be safe to fire and shall reveal no evidence of failure when tested as specified in 4.5.5. Metal parts failure shall be:

a. Loss or break-up of metal parts components during gun launch or flight.

b. Loose or fanning of rotating band.

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c. Shear of rotating band as evidenced by complete lack of engraving.

d. Leakage of propellant gases into base plug thread area.

3.7 First article inspection.-This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.8 Workmanship.-The requirements for workmanship are as shown on the applicable drawings, in the applicable specifications and the following.

3.8.1 Threads.-Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing and shall be in accordance with the provisions of MIL-A-2550.

3.8.2 Burr.-No parts shall have a burr.

3.8.3 Foreign matter.-All parts and assemblies shall be free of chips, dirt, grease, rust, corrosion or other foreign matter.

3.8.4 Rotating band damage.-The rotating band after machining shall be protected from being nicked, burred or otherwise damaged during subsequent handling and preparation for shipment.

3.8.5 Painting and marking.-Painting and marking shall be applied in accordance with the applicable drawings and specifications. The rotating band and threads of the projectile body shall be covered in a manner such as to preclude the application of paint to these surfaces.

4. QUALITY ASSURANCE

4.1 Responsibility for inspection and standard quality assurance provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections.-The following types of inspection shall be performed on this item:

a. First Article Inspection.
b. Quality Conformance Inspection.

4.3 First article inspection

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4.3.1 Submission.-The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation with the provisions of 4.3.2. The first article sample shall consist of the following items in the sample quantities indicated.

Part Description	Drawing	Quantity
Projectile MPTS	10543024	80
Body (Unpainted)	10551955	30
Base Plug	10535957	85

4.3.2 Inspections to be performed.-See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection.-See MIL-A-48078.

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TABLE I

FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
	Projectile, 165MM, TP, M624, Metal Parts		See Below
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Base Plug				
	(Dwg. 10535957)				
	Examination for defects	25		3.1	4.4.2.2
	Mechanical Properties	5		3.3	4.5.1.2
	Salt Spray Test	5		3.4	4.5.4
	Body (Unpainted)				
	(Dwg. 10551955)				
	Examination for Defects	25		3.1	4.4.2.3, 4.4.2.4, 4.4.2.5
	Mechanical Properties	5		3.3	4.5.1.1
	Projectile Body				
	(Dwg. 10543024)				
	Examination for Defects	25		3.1	4.4.2.6
	Salt Spray Test	5		3.4	4.5.4
	Ballistic Test				
	Performance Phase	25		3.6	4.5.5.1, 4.5.5.3
	Over pressure Phase	25		3.6	4.5.5.2, 4.5.5.3

NOTES:

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4.4 Quality conformance inspection

4.4.1 Inspection lot formation.-Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.1.1 Stress relieved lots.-Stress relieved lots shall contain cold worked bodies from not more than one heat of steel that have been cold worked by the same process, utilizing the same tooling and equipment and stress relieved in the same furnace at the same temperature for the same length of time. The stress relieved lot thus constituted shall be the basis for the selection of samples for the mechanical properties test hereinafter prescribed.

4.4.1.2 Projectile lots.-When it has been determined that bodies of a stress relieved lot have complied with the mechanical properties requirement of the applicable drawing, they may be re-grouped for assembly into projectile lots, to which will be assigned a projectile lot number. The projectile lot shall consist of bodies and base plugs produced by one supplier under one contract in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision, under conditions of continuous production free from interruptions other than those due to the end of shift, day, or workweek. The projectile lot thus constituted shall be the basis for the selection of samples for the functional tests hereinafter prescribed.

4.4.2 Examination.-See MIL-A-48078.

a. Sampling plans.-Unless otherwise specified in the Classification of Defects and Tests tables, sampling plans and procedures shall be in accordance with MIL-STD-105, Inspection Level II.

4.4.2.1 Visual inspections.-In the event of a dispute, four power (4X) magnification shall be used in performing visual inspections for defects such as burrs and foreign matter. The roughness comparison specimens prescribed by ANSI - B46.1 shall be used as a basis for surface roughness determination.

4.4.2.1.1 Defect reporting.-The contractor shall prepare a monthly report of all critical defects found, including the number of projectile metal parts inspected and corrective action taken, in accordance with Data Item Description DI-R-1721 and deliver the reports as specified in the contract.

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4.4.2.1.2 Control tool manufacture.-Dimensions marked tool control shall be gaged at the beginning of each eight hours production or fraction thereof from each machine in operation and prior to any tooling change. Where destruction of components is necessary to inspect these dimensions, measurements of the tool may be substituted, provided the contractor has established correlation between the tool dimensions and the component dimensions prior to the start of production. The contractor shall be responsible for maintaining adequate surveillance to assure than drawing requirements are being met throughout the life of the tool. All material produced prior to periodic inspection will be rejected whenever a defective tool control dimension is found.

4.4.2.1.3 Inspection of slug prior to drawing.-Each slug obtained from a billet or bloom shall be visually inspected for surface indications of cracks, splits, pipe, porosity, folds, seams and other metal defects visible to the unaided eye on all exterior surfaces of the slug. Visual standards may be selected to aid in the determination of a slug's acceptability.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50398 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.2	Base Plug		10535957
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Metal defective			3.5.3	Visual
Major					
101	Runout of flange diameter with pitch diameter of thread and front face of flange		0.40%	3.1	Gage
102	True position of recess diameter with pitch diameter of thread and front face of flange		0.40%	3.1	Gage
103	True position of boss diameter with pitch diameter of thread and front face of flange		0.40%	3.1	Gage
104	Pitch diameter of thread, min.		0.40%	3.1	Gage
105	Major diameter of thread, min.		0.40%	3.1	Gage
106	Diameter of flange		0.40%	3.1	Gage
107	Runout of front face of flange with pitch diameter of threads		0.40%	3.1	Gage
108	Depth of recess		0.40%	3.1	Gage
109	Length from rear face to front of flange		0.40%	3.1	Gage
110	Thread damaged or not full		0.40%	3.8.1	Visual
111	Mechanical properties	(a)		3.3	Test/4.5.1.2
112	Salt spray resistance	(b)		3.4	Test/4.5.4

NOTES: a. See 4.4.3.2 for sample size
b. See 4.4.3.3 for sample size

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50398 (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.2	CONTINUATION SHEET				10535957
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Pitch diameter of thread, max.		0.65%	3.1	Gage
202	Diameter of thread chamfer		0.65%	3.1	Gage
203	Diameter of recess		0.65%	3.1	Gage
204	Diameter of boss		0.65%	3.1	Gage
205	Length, total		0.65%	3.1	Gage
206	Thickness of flange		0.65%	3.1	Gage
207	Width of thread undercut		0.65%	3.1	Gage
208	Location of wrench holes at recess		0.65%	3.1	Gage
209	Diameter of wrench holes		0.65%	3.1	Gage
210	Depth to shoulder of recess from rear face		0.65%	3.1	Gage
211	Diameter of thread undercut		0.65%	3.1	Gage
212	Surface finish improper		0.65%	3.1	Gage
213	Radii missing or incorrect		0.65%	3.1	Manual
214	Chamfer missing or incorrect		0.65%	3.1	Visual
215	Protective coating missing or inadequate		0.65%	3.1	Visual
216	Presence of burr		0.65%	3.1	Visual
217	Foreign matter present		0.65%	3.8.2	Manual
218	Marking missing, incorrect or illegible		0.65%	3.8.3	Visual
219	Evidence of poor workmanship		0.65%	3.1	Visual
				3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50398 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.3	Body, Prior to Machining		10551955
			NEXT HIGHER ASSEMBLY 10543024

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Metal defective				Visual
Major					
101	Mechanical properties	a		3.3	Test/4.5.1.1
Minor	None defined				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50398 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.4	Body, Prior to Application of Rotating Band		10551955
			NEXT HIGHER ASSEMBLY 10543024

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of rotating band seat (if applicable)		0.40%	3.1	Gage
102	Length from shoulder to front of band seat (if applicable)		0.40%	3.1	Gage
103	Foreign matter on band seat (if applicable)		0.40%	3.8.3	Visual
Minor					
201	Length from shoulder to rear of rotating band seat (if applicable)		0.65%	3.1	Gage
202	Presence of burr		0.65%	3.8.2	Manual
203	Foreign matter, except as otherwise classified		0.65%	3.8.3	Visual
204	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50398 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 3	DRAWING NUMBER
4.4.2.5	Body, Prior to Painting		10551955
			NEXT HIGHER ASSEMBLY
			10543024

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Hydrostatic test			3.1	Test/4.5.2
2	Air pressure test			3.1	Test/4.5.3
3	Metal defective			3.5.2	Visual
Major					
101	Runout of pitch diameter of external thread with bourrelet and shoulder		0.40%	3.1	Gage
102	Runout of pitch diameter of internal thread with bourrelet and counterbore		0.40%	3.1	Gage
103	True position of counterbore with pitch diameter of internal thread		0.40%	3.1	Gage
104	Runout of rotating band flat with bourrelet		0.40%	3.1	Gage
105	Diameter of bourrelet, min.		0.40%	3.1	Gage
106	Body diameter rear of rotating band		0.40%	3.1	Gage
107	Diameter of rotating band flat		0.40%	3.1	Gage
108	Length from rear face to shoulder		0.40%	3.1	Gage
109	Length from shoulder to rear undercut of rotating band, max. (if applicable)		0.40%	3.1	Gage
110	Length from shoulder to front undercut of rotating band, min. (if applicable)		0.40%	3.1	Gage

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MIL-P-50398 (AR)

PARAGRAPH	TITLE	SHEET 2 OF 3	DRAWING NUMBER
4.4.2.5	CONTINUATION SHEET		10551955
			NEXT HIGHER ASSEMBLY 10543024

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
111	Length from shoulder to forward edge of rotating band flat		0.40%	3.1	Gage
112	Thickness of wall at nose		0.40%	3.1	Gage
113	Thickness of wall at datum length (from ogive)		0.40%	3.1	Gage
114	Thickness of wall at rear datum, min.		0.40%	3.1	Gage
115	Variation in wall thickness improper		0.40%	3.1	Gage
116	Width of rotating band flat		0.40%	3.1	Gage
117	Profile of rotating band		0.40%	3.1	Gage
118	Profile of nose (after hydrostatic and air tests)		0.40%	3.1	Gage
119	Pitch diameter of internal thread, max.		0.40%	3.1	Gage
120	Minor diameter of internal thread, max.		0.40%	3.1	Gage
121	Pitch diameter of external thread, min.		0.40%	3.1	Gage
122	Major diameter of external thread, min.		0.40%	3.1	Gage
123	Length of effective internal thread, min.		0.40%	3.1	Gage
124	Thread(s) damaged or not full		0.40%	3.8.1	Visual
Minor					
201	Pitch diameter of internal thread, min.		0.65%	3.1	Gage
202	Pitch diameter of external thread, max.		0.65%	3.1	Gage
203	Length of internal thread, max.		0.65%	3.1	Gage
204	Diameter of counterbore		0.65%	3.1	Gage
205	Diameter of internal thread chamfer		0.65%	3.1	Gage
206	Diameter of external thread chamfer		0.65%	3.1	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50398 (AR)

PARAGRAPH	TITLE	SHEET 3 OF 3	DRAWING NUMBER
4.4.2.5	CONTINUATION SHEET		10551955
			NEXT HIGHER ASSEMBLY
			10543024

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
207	Diameter of external thread undercut		0.65%	3.1	Gage
208	Diameter of cavity forward of internal thread		0.65%	3.1	Gage
209	Depth to bottom of counterbore		0.65%	3.1	Gage
210	Length, total		0.65%	3.1	Gage
211	Width of external thread undercut		0.65%	3.1	Gage
212	Depth of undercut front and rear of rotating band (if applicable)		0.65%	3.1	Gage
213	Width of undercut front and rear of rotating band (if applicable)		0.65%	3.1	Gage
214	Surface finish improper		0.65%	3.1	Manual
215	Foreign matter present		0.65%	3.8.3	Visual
216	Presence of burr		0.65%	3.8.2	Manual
217	Radii missing or incorrect		0.65%	3.1	Visual
218	Chamfer missing or incorrect		0.65%	3.1	Visual
219	Marking missing, incorrect or illegible		0.65%	3.1	Visual
220	Evidence of poor workmanship		0.65%	3.8	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50398 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.6	Projectile, 165MM, TP, M624, Metal Parts				10543024
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Metal defective			3.5.2	Visual
2	Any required inspection acceptance stamp missing			3.1	Visual
Major					
101	Diameter of bourrelet, max.		100%	3.1	Gage
102	Weight		0.40%	3.1	Balance
103	Salt spray resistance	a		3.4	Test/4.5.4
104	Ballistic test	b		3.6	Test/4.5.5
Minor					
201	Paint on threads		0.65%	3.8	Visual
202	Protective coating missing or inadequate		0.65%	3.4	Visual
203	Paint on rotating band bearing surface		0.65%	3.8	Visual
204	Foreign matter present		0.65%	3.8.3	Visual
205	Presence of burr		0.65%	3.8.2	Manual
206	Protective cap missing		0.65%	3.1	Visual
207	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES: a. See 4.4.3.3 for sample size.
b. See 4.4.3.4 for sample size.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50398 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.7	Carton, Packing, Prior to Sealing				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Any support missing or damaged		0.65%	3.1	Visual
202	Separator(s) missing or damaged		0.65%	3.1	Visual
203	Contents improperly packed		0.65%	3.1	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50398 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.8	Cartons, Packed, After Sealing				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Contents loose		0.65%	3.1	Manual
202	Carton damaged, cut or gouged through all layers		0.65%	3.1	Visual
203	Carton not properly sealed		0.65%	3.1	Visual
204	Marking missing, incorrect or illegible		0.65%	3.1	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-50398 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.9	Pallet				8837835
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Strapping missing, broken or loose		0.65%	3.1	Visual/Manual
202	Pallet damaged		0.65%	3.1	Visual
203	Improperly packed		0.65%	3.1	Visual
204	Marking missing, improper or illegible		0.65%	3.1	Visual

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.4.3 Testing

4.4.3.1 Tensile strength of projectile body.-Two projectile bodies from the first batch of a stress-relieved lot shall be subjected to this test. Following satisfactory compliance of the initial sample with the requirements of dwg. 10551955, two projectile bodies from every five stress-relieved batches thereafter shall be selected for this test. Failure of any tensile strength specimen to meet the yield strength and elongation requirements of dwg. 10551955 shall reject the batch of stress-relieved bodies which that specimen represents, and shall require the establishment of a new stress-relief lot as prescribed herein.

4.4.3.2 Tensile strength of base plug.-Two base plugs from the first batch of a stress-relief lot of plugs shall be tested for compliance with the yield strength and elongation requirements specified on dwg. 10535957. Following the establishment of a stress-relieved lot of plugs, two base plugs from every five batches shall be subjected to this test. Failure of any base plug tensile specimen to comply with the requirements of dwg. 10535957 shall reject the batch of base plugs represented by the specimen and shall require the establishment of a new stress-relieved base plug lot. If stress-relieving or heat treatment is not performed, two base plugs per five thousand plugs shall be subjected to this test.

4.4.3.3 Surface preparation and application of protective finish.-The sample size and acceptance criteria for the inspection of surface preparation and the application of protective finish shall be in accordance with the applicable specifications. Failure to comply with the applicable specification and drawing requirements shall be cause for taking the corrective action recommended therein (see 4.5.4).

4.4.3.4 Proving ground tests.-The following tests shall be performed as specified in 4.5.5. Any sample failing the safety criteria specified in 3.6 shall reject the lot represented by that sample.

4.4.3.4.1 Performance phase.-Beginning with the first three lots of projectile metal parts, twenty five samples shall be fired for performance and safety. Sampling shall then revert to ten samples per lot thereafter.

4.4.3.4.2 Overpressure phase.-Beginning with the first three lots of projectile metal parts, twenty-five samples shall be fired at overpressure for security and safety. Sampling shall revert to ten samples per lot thereafter.

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4.4.4 Inspection equipment.-The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with 4.4.2.2. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.2 herein.

4.5 Test methods and procedures

4.5.1 Mechanical properties.-The criteria for obtaining specimens and performing these tests shall be as prescribed by ASTM E8, except that elongation shall be measured to the nearest thousandth (0.000) prior to conversion into percent elongation. Percent elongation shall be recorded to the nearest tenth (0.0) percent. The number of specimens and their location are for the convenience of the contractor and Government, and shall not restrict the contractor or Government from testing additional specimens at other locations in order to determine that the mechanical properties of stress-relieved bodies or plugs meet the drawing requirements.

4.5.1.1 Tensile test of projectile body.-From the samples provided in accordance with 4.4.3.1, two tensile test specimens conforming to the largest obtainable standard specimen for tubular products, as prescribed by ASTM E8, shall be taken from opposite sides of the body at the rear of the projectile body.

4.5.1.2 Tensile test of base plug.-From the samples provided in accordance with 4.4.3.2, one tensile test specimen conforming to the largest obtainable standard specimen, as prescribed by ASTM E8, shall be taken from each base plug.

4.5.2 Hydrostatic test.-The projectile body assembly shall be hydrostatically pressure tested using equipment capable of performing the test accurately and consistently. The body shall be mounted in an approved test fixture complying with the requirements of dwg. 10551955 such that three inches of the nose remains unsupported. In performing the test, the hydrostatic pressure will be applied gradually until the required pressure is reached, after which the pressure will be held for thirty seconds, minimum. The test equipment must be capable of stamping an acceptance mark (H) for accepted bodies in the location required by dwg. 10551955. A fail-safe system of operation shall also be required (see 6.3).

4.5.2.1 Calibration.-The hydrostatic and air pressure test equipment (see 4.5.3.1) shall be calibrated prior to the start of each day's operation and at periods not exceeding four hours of continuous operation. It shall also be calibrated prior to the restarting of the operation after a lay-off of two hours of more; or, following the completion of unscheduled repairs or

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maintainance. The contractor shall prepare a written standard operating procedure, to include the inspection of projectile bodies and calibration procedures, which must be reviewed and approved by the Contracting Officer prior to the submission of the first article sample (see 6.2).

4.5.3 Air pressure test.-The projectile body shall be subjected to this test using equipment capable of performing this test accurately and consistently. Prior to testing, each projectile shall be inspected for the presence of the hydrostatic pressure acceptance stamp and each projectile body shall be dried in accordance with the requirement of dwg. 10551955. The body shall be mounted in an approved fixture in a manner such that the body shall be entirely submerged in water following its pressurization. Observation shall be made for air leakage from each projectile body, especially the nose area. In performing this test, the air pressure will be applied until the required air pressure is reached, after which the air pressure will be held for fifteen seconds, minimum. The test equipment must be capable of stamping an acceptance mark (A) for accepted bodies in the location required by dwg. 10551955. A fail-safe system of operation shall also be required (see 6.3)

4.5.3.1 Calibration.-The air pressure equipment shall be calibrated in accordance with the requirements of 4.5.2.1. The contractor shall prepare a written standard operating procedure, to include the inspection of projectile bodies and calibration procedures, which must be reviewed and approved by the Contracting Officer prior to the submission of the first article sample (see 6.2).

4.5.4 Surface preparation and application of protective finish.-The procedures for testing the cleaning and surface treatment methods for components and assemblies shall be in accordance with the methods and acceptance criteria for testing as specified in the applicable specification. A salt spray (fog) test shall be performed on projectile bodies and base plugs, in accordance with the requirements of the applicable drawings and specifications, using the equipment and test procedures prescribed by ASTM B117.

4.5.5 Proving ground test.-The samples which are to be ballistically tested as prescribed herein shall be fired in a M135 gun. The ammunition in each phase shall be temperature conditioned to $70^{\circ} \pm 2.5^{\circ}F$ ($21^{\circ} \pm 1.5^{\circ}C$) until all propellant is stabilized at a uniform temperature prior to firing, which normally requires a period of twenty-four hours, minimum. Adequate photographic equipment shall be used to assure satisfactory coverage of each test sample's integrity and flight from the gun muzzle to the first one hundred feet of projectile flight, minimum. Muzzle velocity and chamber pressure shall be recorded for all rounds fired (see 6.4).

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4.5.5.1 Performance Phase.-The samples selected for this phase will be fired using the service velocity propelling charge in a gun tube with a fifty percent remaining life, minimum. Target impact dispersion data shall be taken on a vertical cloth target for information only. The target distance is to be 900 meters from the muzzle of the gun and the target size is to be six meters by six meters, minimum. Spotter rounds are to be used to assure that the weapon is zeroed and aimed at the center of the target. An aiming point or stake may be used for this purpose. Target impacts, including spotters, shall be measured to the nearest one-half inch in both the horizontal and vertical distance from target center. The horizontal and vertical hit pattern results shall be expressed as probable errors in mils and shall be recorded on the firing record. Target miss coordinates will be estimated and recorded along with pertinent flight observations.

4.5.5.2 Overpressure phase.-The samples selected for this phase will be fired, using an adjusted propelling charge to obtain a minimum mean pressure of 5600 pounds per square inch (psi) and a maximum individual pressure not exceeding 6000 psi, in a gun tube with twenty-five percent remaining life, maximum.

4.5.5.3 Safety.-All samples fired in proving ground tests shall be observed for safety, in particular, the breakup of projectile metal parts. Any ballistic sample failing the safety criteria specified herein shall be cause for the cessation of the test phase in which the failure occurs, as well as, the remainder of the ballistic test and shall reject the projectile metal parts lot represented by that sample.

4.5.5.4 Assembly of test sample cartridges.-The projectile metal parts samples shall be assembled into cartridges in accordance with dwg. 9219045 and MIL-C-50413.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.-Packing and marking shall be Level B in accordance with MIL-STD-1169 and this specification.

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5.2 Packing

5.2.1 Body assembly.-The projectile body with protector cap shall be packaged in accordance with dwg. 8837835.

5.2.2 Base plug.-Ten base plugs shall be packed in a fiberboard box, conforming to the requirements of PPP-B-636, Style RSC. Each base plug shall be separated by a fiberboard disc. Additional discs shall be added, as required, to make a tight pack.

5.2.2.1 Material.-Fiberboard boxes and discs shall meet the requirements of PPP-F-320, Type CF, Domestic Class, Variety SW, Grade 200, B Flute.

5.2.2.2 Size.-Dimensions of boxes shall be 5 1/2 X 5 1/2 X 12 3/16 inches deep. Dimension of the disc shall be 5.0 inches in diameter.

5.3 Shipping.-When component parts from more than one lot are shipped as a carload, each lot must be kept separate and the division between lots shall be clearly indicated to prevent mixing of lots while in transit.

5.4 Marking

5.4.1 Body assembly.-Palletized body assemblies shall be marked in accordance with dwg. 8837835.

5.4.2 Base plug.-Packaged base plugs shall be marked, Level B, in accordance with MIL-STD-1169.

6. NOTES

6.1 Ordering data

6.1.1 Procurement requirements.-See MIL-A-48078.

6.1.2 Data requirements.-Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
4.4.2.1.1	Quality Inspection Defect Report	DI-R-1721

(Copies of data item descriptions required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

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6.1.2.1 Delivery of data to the technical activity.-The following address is to be added to those listed in the contract (Form DD 1423) for delivery of one copy, each, to the technical activity, of the Ammunition Data Cards and Quality Inspection Defect Reports prepared by the contractor for the duration of the contract: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-E, Dover, New Jersey 07801.

6.2 Submission of designs for approval.-See paragraph 6.2.3 of MIL-A-48078. Submit equipment designs, as required, to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-I, Dover, New Jersey 07801.

6.3 Failsafe.-An inspection system required to be fail-safe is defined as a system in which the failure of any electrical, mechanical, hydraulic, or pneumatic portion of the system shall result in the placement of the system in the reject mode of operation and shall require a definite signal to accept the item being inspected.

6.4 Firing record information.-The ballistic test firing records shall include the serial numbers of all weapons and fire control components. Stock component lot numbers of warmers and/or spotters shall also be included. For each round fired, the record shall contain deflection, elevation, time of firing, chamber pressure, pertinent flight characteristics, and the results of the examination of the smear photos from the overpressure phase. The following addresses are to be added to the Ballistic Test Request (BTR) for distribution of firing test reports to the technical activity: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-E and DRDAR-LCU-CV, Dover, New Jersey 07801.

6.5 Visual standards.-Visual standards for metal defective discontinuities shall be submitted to the Contracting Officer for approval prior to their implementation in the remainder of the contract in accordance with the provisions prescribed herein and in the contract. The resubmission of visual standards for approval on subsequent contract will also be required. The contractor should be cognizant of the following criteria which the Government will use as part of its determination in judging the acceptance of each proposed standard for subsequent use:

a. There shall be no degradation of the physical and mechanical properties in the area surrounding the discontinuity;
b. Discontinuities should not be of a depth greater than ten percent of the wall thickness at the location of the discontinuity.

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6.6 Proving ground test summary

Phase	Sample Size First Article	Production	Requirement	Sampling	Procedure
Performance	25	10*	Safety TID (Info)	4.4.3.4.1	4.5.5.3 4.5.5.1
Over-pressure	25	10*	Safety Recovery	4.4.3.4.2	4.5.5.3 4.5.5.2

*Sample size for first three lots shall be twenty-five projectile metal parts.

Custodian:
ARMY-AR

Preparing Activity:
ARMY-AR

Project Number: 1320-A495

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-P-50398 PROJECTILE, 165MM, TP M624 Metal Parts

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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